ENUNCIADO — PROBLEMA 1

Math 685: Preliminary Exam – Spring 2014

JANUARY 15

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   (a) Formulate a least squares problem for the polynomial fit of the form $Ay = b$. Explain how you will define the entries of the matrix A and the vector b.

   (b) Derive a formula for the solution of the least squares problem.

2. *Gaussian Quadrature.*

   (a) Derive the two point Gaussian integration formula $G_2(f)$ for

   $$I(f) = \int_{-1}^{1} f(x)dx.$$

   Indicate for what polynomial degree this rule is exact.

   (b) Evaluate the integral using the two-point Gauss Quadrature fourmula derived:

   $$\int_1^4 \frac{dx}{x+2}.$$

3. Suppose you need to minimize $f(x) = \frac{1}{2}x^T Ax - b^T x + c$, $x \in \mathbb{R}^n$, where A is symmetric positive definite.

   (a) Let x_0 be the starting point. Perform one step of the steepest descent method with exact line search and find the expression for x_1.

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   (d) Compare and contrast in general terms the Jacobi method and the Gauss-Seidel iterative methods.

4. Given a matrix $S = [s_1, \ldots, s_N] \in \mathbb{R}^{M \times N}$ of rank $K \leq N$ and $M \geq N$, the *proper orthogonal decomposition (POD)* is a method to approximate S with $d \ll K$ linearly independent vectors in $\text{span}\{s_i\}_{i=1}^N$. Let $S = U\Sigma V^T$ be the singular value decomposition of S, i.e.

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   are orthogonal matrices and $\Sigma = \begin{bmatrix} D \\ 0 \end{bmatrix} \in \mathbb{R}^{M \times N}$, with diagonal matrix $D = \text{diag}(\sigma_1, \ldots, \sigma_N)$ containing the singular values of S:

   $$\sigma_1 \geq \cdots \geq \sigma_K > \sigma_{K+1} = \cdots = \sigma_N = 0.$$

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SOLUCIÓN — PROBLEMA 1

Problema 1 — Ajuste polinomial por mínimos cuadrados

(a) Formulación del problema $Ay = b$.

Se tienen $n+1$ datos (x_j, f_j), $j = 0, \dots, n$, con $x_j = j/n \in [0,1]$, y se busca el polinomio de grado $k \ll n$

$$p(x) = \sum_{i=0}^{k} y_i \, x^i$$

que mejor aproxima los datos en el sentido de mínimos cuadrados. Como hay $n+1$ condiciones $p(x_j) \approx f_j$ y solo $k+1$ incógnitas $y_0, \dots, y_k$ (con $k+1 \le n+1$, en general $k+1 < n+1$), el sistema

$$p(x_j) = \sum_{i=0}^{k} y_i x_j^i = f_j, \qquad j = 0, \dots, n,$$

está **sobredeterminado**. En forma matricial $Ay = b$ con

$$A = \left(a_{ji}\right) \in \mathbb{R}^{(n+1)\times(k+1)}, \quad a_{ji} = x_j^{\,i} \;\; (j = 0,\dots,n,\; i = 0,\dots,k), \qquad b = (f_0, f_1, \dots, f_n)^T \in \mathbb{R}^{n+1}.$$

A es una matriz tipo Vandermonde rectangular (alta y delgada, pues $k+1 \ll n+1$). Se busca $y \in \mathbb{R}^{k+1}$ que minimice $\|Ay - b\|_2^2$, pues en general el sistema no tiene solución exacta.

(b) Solución del problema de mínimos cuadrados.

Rango completo de A. Si $Ay = 0$ con $y \neq 0$, entonces el polinomio $q(x) = \sum_{i=0}^{k} y_i x^i$ (de grado $\le k$, no idénticamente nulo) se anularía en los $n+1$ puntos distintos $x_0, \dots, x_n$. Pero un polinomio no nulo de grado $\le k$ tiene a lo más k raíces, y $n+1 > k$ (pues $k \ll n$). Contradicción. Luego A tiene rango columna completo $k+1$, y $A^TA \in \mathbb{R}^{(k+1)\times(k+1)}$ es invertible (de hecho SPD).

Ecuaciones normales. Sea \$\phi(y)=\|Ay-b\|_2^2=(Ay-b)^T(Ay-b) =y^TA^TAy-2y^TA^Tb+b^Tb $.Como$\phi\$ es cuadrática convexa (su Hessiano es $2A^TA \succeq 0$), su mínimo se alcanza donde el gradiente se anula:

$$\nabla_y \phi(y) = 2A^TAy - 2A^Tb = 0 \implies A^TAy = A^Tb.$$

Por la invertibilidad de A^TA:

$$\boxed{y = (A^TA)^{-1}A^Tb.}$$

Verificación dimensional: $A^T \in \mathbb{R}^{(k+1)\times(n+1)}$, así $A^TA \in \mathbb{R}^{(k+1)\times(k+1)}$ y $A^Tb \in \mathbb{R}^{k+1}$: la fórmula produce efectivamente un vector $y \in \mathbb{R}^{k+1}$, como se requiere.

Nota numérica. En la práctica, para k moderado, resolver $A^T Ay = A^T b$ directamente puede estar mal condicionado ($\kappa(A^T A) = \kappa(A)^2$); es preferible usar la factorización $A = QR$ (Householder o Gram-Schmidt) y resolver el sistema triangular $R_1 y = Q_1^T b$ (ver Problema 4(b) del examen de Agosto 2011), o emplear una base de polinomios ortogonales discretos en vez de la base monomial.

ENUNCIADO — PROBLEMA 2

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SOLUCIÓN — PROBLEMA 2

Problema 2 — Cuadratura gaussiana de dos puntos

(a) Derivación de $G_2(f)$ para $I(f) = \int_{-1}^{1} f(x)\ dx$.

Se buscan nodos x_1, x_2 y pesos w_1, w_2 tales que

$$G_2(f) = w_1 f(x_1) + w_2 f(x_2)$$

sea exacta para todo polinomio de grado ≤ 3 (4 parámetros libres $\Rightarrow$ 4 condiciones de exactitud, para $f = 1, x, x^2, x^3$). Por la simetría del intervalo $[-1, 1]$, se busca una solución simétrica $x_1 = -x_2 =: t > 0$, $w_1 = w_2 =: w$.

- $f(x) = 1$: $\displaystyle\int_{-1}^{1} 1\ dx = 2 = w + w = 2w\ \Rightarrow\ w = 1.$
- $f(x) = x$: $\displaystyle\int_{-1}^{1} x\ dx = 0 = w(t) + w(-t) = 0$, automáticamente

satisfecha por la simetría.

- $f(x) = x^2$: $\displaystyle\int_{-1}^1 x^2dx=\frac23 = wt^2+wt^2=2t^2

\ \Rightarrow\ t^2=\frac13\ \Rightarrow\ t=\frac{1}{\sqrt3}.$

- $f(x) = x^3$: $\displaystyle\int_{-1}^{1} x^3 dx = 0 = wt^3 + w(-t)^3 = 0$, automáticamente

satisfecha (impar).

Luego

$$\boxed{G_2(f) = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right),}$$

que es **exacta para polinomios de grado ≤ 3**. Esto coincide con la cota general de Gauss: una regla de n nodos es exacta a lo más de grado $2n - 1$, y aquí se alcanza esa cota ($2 \cdot 2 - 1 = 3$) porque los nodos $\pm 1/\sqrt{3}$ son las raíces del polinomio de Legendre $P_2(x) = \frac{1}{2}(3x^2 - 1)$, como exige la teoría general de cuadratura gaussiana.

(b) Evaluación de $\displaystyle\int_{1}^{4} \frac{dx}{x+2}$.

Cambio de variable lineal de $[1, 4]$ a $[-1, 1]$: $x = \dfrac{3}{2}t + \dfrac{5}{2}$, $dx = \dfrac{3}{2}\ dt$. Entonces

$$\int_{1}^{4} \frac{dx}{x+2} = \int_{-1}^{1} \frac{\frac{3}{2}\,dt}{\frac{3}{2}t + \frac{5}{2} + 2} = \int_{-1}^{1} \frac{\frac{3}{2}\,dt}{\frac{3}{2}(t+3)} = \int_{-1}^{1} \frac{dt}{t+3}.$$

Aplicando G_2 con $t = \pm 1/\sqrt{3}$:

$$G_2 = \frac{1}{3+\frac{1}{\sqrt{3}}} + \frac{1}{3-\frac{1}{\sqrt{3}}} = \frac{\left(3-\frac{1}{\sqrt{3}}\right)+\left(3+\frac{1}{\sqrt{3}}\right)}{9-\frac{1}{3}} = \frac{6}{26/3} = \frac{18}{26} = \boxed{\frac{9}{13} \approx 0.69231}.$$

Verificación: el valor exacto es $\int_1^4 \frac{dx}{x+2} = \ln(x+2)\big|_1^4 = \ln 6 - \ln 3 = \ln 2 \approx 0.69315$. La diferencia $|9/13 - \ln 2| \approx 0.0008$ es pequeña y consistente con el error de cuadratura gaussiana $E = \frac{f^{(4)}(\xi)}{135}$ para $n = 2$ (aquí $f(x) = 1/(x+3)$ en $[-1, 1]$ tiene cuarta derivada acotada, dando un error del orden correcto).

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SOLUCIÓN — PROBLEMA 3

Problema 3 — Minimización de una forma cuadrática: descenso de gradiente y método de Jacobi

Sea $f(x) = \frac{1}{2}x^T Ax - b^T x + c$, $A \in \mathbb{R}^{n\times n}$ simétrica definida positiva (SPD).

(a) Un paso de descenso de gradiente (steepest descent) con búsqueda de línea exacta.

El gradiente es $\nabla f(x) = Ax - b$ (válido porque A es simétrica). La dirección de descenso más pronunciado en x_0 es $-\nabla f(x_0) = b - Ax_0 =: r_0$ (el *residuo*). Se busca $x_1 = x_0 + \alpha_0 r_0$ con α_0 que minimiza $\varphi(\alpha) := f(x_0 + \alpha r_0)$. Derivando:

$$\varphi'(\alpha) = r_0^T A(x_0 + \alpha r_0) - b^T r_0 = r_0^T (Ax_0 - b) + \alpha r_0^T Ar_0 = -r_0^T r_0 + \alpha\, r_0^T Ar_0.$$

Igualando a cero (y usando $r_0^T Ar_0 > 0$ pues A SPD y $r_0 \neq 0$ en general):

$$\alpha_0 = \frac{r_0^T r_0}{r_0^T Ar_0}.$$

Por lo tanto

$$\boxed{x_1 = x_0 + \frac{r_0^T r_0}{r_0^T Ar_0}\, r_0, \qquad r_0 = b - Ax_0.}$$

(b) f se minimiza cuando $Ax = b$.

Sea $x^* = A^{-1}b$ (existe pues A SPD $\Rightarrow$ invertible). Para cualquier $x \in \mathbb{R}^n$, escribamos $x = x^* + e$. Entonces, usando $Ax^* = b$ y la simetría de A (de modo que $x^{*T} Ae = (Ax^*)^T e = b^T e$):

$$f(x^* + e) = \tfrac{1}{2}(x^* + e)^T A(x^* + e) - b^T (x^* + e) + c$$
$$= \underbrace{\tfrac{1}{2}x^{*T} Ax^* - b^T x^* + c}_{=f(x^*)} + \underbrace{\left(x^{*T} Ae - b^T e\right)}_{=0} + \tfrac{1}{2}e^T Ae = f(x^*) + \tfrac{1}{2}e^T Ae.$$

Como A es SPD, $\frac{1}{2}e^T Ae \geq 0$ con igualdad si y solo si $e = 0$. Luego

$$f(x) = f(x^*) + \tfrac{1}{2}(x - x^*)^T A(x - x^*) \;\geq\; f(x^*) \qquad \forall x,$$

con igualdad únicamente en $x = x^* = A^{-1}b$, es decir, exactamente cuando $Ax = b$. Así f tiene un único mínimo global, alcanzado precisamente en la solución del sistema lineal. ■

(c) Primera iterada de Jacobi.

Descomponga $A = D - L - U$ (D diagonal, $-L$ estrictamente triangular inferior, $-U$ estrictamente superior). La iteración de Jacobi es $Dx^{(k+1)} = (L + U)x^{(k)} + b$, es decir

$$x^{(k+1)} = D^{-1}\Big(b - (A - D)x^{(k)}\Big) = x^{(k)} + D^{-1}\Big(b - Ax^{(k)}\Big).$$

Con x_0 el punto inicial y $r_0 = b - Ax_0$:

$$\boxed{x_1 = x_0 + D^{-1} r_0, \qquad D = \operatorname{diag}(a_{11}, \dots, a_{nn}),}$$

o, componente a componente,

$$(x_1)_i = \frac{1}{a_{ii}}\Big(b_i - \sum_{j \neq i} a_{ij}(x_0)_j\Big), \qquad i = 1, \dots, n.$$

(d) Jacobi vs. Gauss-Seidel.

- **Jacobi:** todas las componentes de $x^{(k+1)}$ se calculan usando *únicamente*

valores de la iterada anterior $x^{(k)}$; las actualizaciones son simultáneas/independientes, lo que hace el método trivialmente paralelizable (cada componente se puede calcular en un procesador distinto) y requiere almacenar dos vectores ($x^{(k)}$ y $x^{(k+1)}$).

- **Gauss-Seidel:** al calcular $(x^{(k+1)})_i$ se usan los valores *ya

actualizados* $(x^{(k+1)})_j$ para $j < i$ y los antiguos $(x^{(k)})_j$ para $j > i$; el método es intrínsecamente secuencial (difícil de paralelizar directamente) pero permite sobrescribir el vector en el mismo arreglo (ahorra memoria) y, para clases amplias de matrices (SPD, diagonalmente dominantes), **converge más rápido** que Jacobi — de hecho, para matrices tridiagonales consistentemente ordenadas se cumple la relación clásica $\rho(B_{GS}) = \rho(B_J)^2$, por lo que Gauss-Seidel requiere aproximadamente la mitad de iteraciones que Jacobi para la misma tolerancia. Ambos son métodos iterativos estacionarios de costo $O(n^2)$ por iteración (o $O(\mathrm{nnz})$ para matrices dispersas), y ambos convergen para todo $x^{(0)}$ si y solo si el radio espectral de su respectiva matriz de iteración es menor que 1 (garantizado, por ejemplo, si A es SPD o estrictamente diagonalmente dominante).

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(a) Show that $\text{span}\{s_j\}_{j=1}^N = \text{span}\{u_j\}_{j=1}^K$.

(b) Let $A_i = u_i v_i^T$ be a rank-1 matrix for $1 \le i \le N$. Show the orthogonal decomposition $S = \sum_{i=1}^K \sigma_i A_i$ in the Frobenius inner product

$$\langle A, B\rangle_F = \sum_{i=1}^M \sum_{j=1}^N a_{ij} b_{ij},$$

i.e. show that $\langle A_i, A_j\rangle_F = \delta_{ij}$, where δ_{ij} is the Kronecker delta.

(c) For $d < K$ we may approximate S with $\sum_{i=1}^d \sigma_i A_i$; this is the POD of S. Show the error formula

$$\left\| S - \sum_{i=1}^d \sigma_i A_i \right\|_F^2 = \sum_{i=d+1}^K \sigma_i^2,$$

where $\|\cdot\|_F$ is the norm subordinate to the Frobenius inner product. This expression could be used to find the number d so that the error $\sum_{i=d+1}^K \sigma_i^2$ is within a given tolerance.

5. Consider the two-point boundary value problem

$$\mathcal{A}u := -(a(x)u')' + c(x)u = f(x) \quad \text{in } (0,1), \quad u(0) = u(1) = 0,$$

where $a(x) \ge a_0 > 0$ and $c(x) \ge 0$ are smooth functions in $[0,1]$, and $f(x) \in L^2(0,1)$.

(a) Derive the variational form for this problem.

(b) Set $a(x) = 1$ and $c(x) = 0$ and consider a partition of $[0,1]$

$$0 = x_0 < x_1 < \cdots < x_N = 1.$$

Write the system of equations produced by the finite element method with continuous piecewise linear elements $\Phi_j(x)$, $j = 1, 2, \ldots, N-1$,

$$\Phi_j(x) = \begin{cases} \frac{x - x_{j-1}}{x_j - x_{j-1}}, & x_{j-1} \le x \le x_j, \\ \frac{x_{j+1} - x}{x_{j+1} - x_j}, & x_j \le x \le x_{j+1}, \\ 0, & \text{otherwise.} \end{cases}$$

(c) Show that this system of equations is equivalent to the one produced by the finite difference method for the uniform partition of $f \equiv$ constant.

2

SOLUCIÓN — PROBLEMA 4

Problema 4 — Descomposición en valores singulares y POD

Sea $S = U\Sigma V^T \in \mathbb{R}^{M\times N}$, $\operatorname{rank}(S) = K \le N \le M$, con $\sigma_1 \ge \cdots \ge \sigma_K > \sigma_{K+1} = \cdots = \sigma_N = 0$. Como $\sigma_i = 0$ para $i > K$, la SVD se puede escribir como suma de K términos de rango 1:

$$S = \sum_{i=1}^{K} \sigma_i \, u_i v_i^T. \qquad (*)$$

(a) $\operatorname{span}\{s_j\}_{j=1}^N = \operatorname{span}\{u_j\}_{j=1}^K$.

(⊆) De $(*)$, la columna j de S es

$$s_j = Se_j = \sum_{i=1}^{K} \sigma_i u_i (v_i)_j = \sum_{i=1}^{K} \left[\sigma_i (v_i)_j\right] u_i,$$

combinación lineal de $u_1, \dots, u_K$. Luego $s_j \in \operatorname{span}\{u_i\}_{i=1}^K$ para todo j, y $\operatorname{span}\{s_j\}_{j=1}^N \subseteq \operatorname{span}\{u_i\}_{i=1}^K$.

(⊇) Para $i \le K$, usando que $V^T v_i = e_i$ (columnas de V ortonormales) y que $\Sigma e_i = \sigma_i \hat{e}_i$ (con $\hat{e}_i \in \mathbb{R}^M$ el i-ésimo vector canónico, ya que $\sigma_i \ne 0$):

$$Sv_i = U\Sigma V^T v_i = U\Sigma e_i = U(\sigma_i \hat{e}_i) = \sigma_i u_i.$$

Como $\sigma_i \ne 0$ (pues $i \le K$),

$$u_i = \frac{1}{\sigma_i} S v_i = \frac{1}{\sigma_i} \sum_{j=1}^{N} (v_i)_j \, s_j \in \operatorname{span}\{s_j\}_{j=1}^N.$$

Esto vale para todo $i = 1, \dots, K$, así que $\operatorname{span}\{u_i\}_{i=1}^K \subseteq \operatorname{span}\{s_j\}_{j=1}^N$.

Combinando ambas inclusiones: $\operatorname{span}\{s_j\}_{j=1}^N = \operatorname{span}\{u_j\}_{j=1}^K$. ■

(b) Ortogonalidad de $A_i = u_i v_i^T$ en el producto de Frobenius.

Para $1 \le i, j \le N$:

$$A_i^T A_j = (u_i v_i^T)^T (u_j v_j^T) = v_i u_i^T u_j v_j^T.$$

Como $u_i^T u_j$ es un escalar, y usando la propiedad cíclica de la traza junto con $\operatorname{tr}(v_i v_j^T) = v_j^T v_i$ (traza de un producto exterior es el producto interior):

$$\langle A_i, A_j \rangle_F = \operatorname{tr}(A_i^T A_j) = \operatorname{tr}\Big(v_i (u_i^T u_j) v_j^T\Big) = (u_i^T u_j)\operatorname{tr}(v_i v_j^T) = (u_i^T u_j)(v_i^T v_j).$$

Como U y V son matrices ortogonales, sus columnas son ortonormales: $u_i^T u_j = \delta_{ij}$ y $v_i^T v_j = \delta_{ij}$. Luego

$$\langle A_i, A_j \rangle_F = \delta_{ij} \cdot \delta_{ij} = \delta_{ij}$$

(pues $\delta_{ij}^2 = \delta_{ij} \in \{0, 1\}$). ■ Esto confirma que $(*)$, $S = \sum_{i=1}^K \sigma_i A_i$, es una descomposición de S en términos **ortonormales** $\{A_i\}$ respecto al producto de Frobenius.

(c) Fórmula del error de la aproximación POD.

Para $d < K$, usando $(\,*)$:

$$S - \sum_{i=1}^{d} \sigma_i A_i = \sum_{i=1}^{K} \sigma_i A_i - \sum_{i=1}^{d} \sigma_i A_i = \sum_{i=d+1}^{K} \sigma_i A_i.$$

Por bilinealidad del producto de Frobenius y la ortonormalidad probada en (b):

$$\left\| \sum_{i=d+1}^{K} \sigma_i A_i \right\|_F^2 = \left\langle \sum_{i=d+1}^{K} \sigma_i A_i,\ \ \sum_{j=d+1}^{K} \sigma_j A_j \right\rangle_F = \sum_{i=d+1}^{K} \sum_{j=d+1}^{K} \sigma_i \sigma_j \underbrace{\langle A_i, A_j \rangle_F}_{=\delta_{ij}} = \sum_{i=d+1}^{K} \sigma_i^2.$$

Es decir,

$$\boxed{\left\| S - \sum_{i=1}^{d} \sigma_i A_i \right\|_F^2 = \sum_{i=d+1}^{K} \sigma_i^2.} \qquad \blacksquare$$

Verificación (caso límite $d = K$): la fórmula da error 0, correcto, pues $\sum_{i=1}^{K} \sigma_i A_i = S$ exactamente. Para $d = 0$ (aproximar por la matriz nula), da $\sum_{i=1}^{K} \sigma_i^2 = \|S\|_F^2$, lo cual también es consistente (identidad de Parseval para la SVD). Esta fórmula (el teorema de Eckart–Young en su versión de energía) permite elegir el rango d de truncamiento de la POD de modo que $\sum_{i=d+1}^{K} \sigma_i^2$ esté por debajo de una tolerancia dada.

ENUNCIADO — PROBLEMA 5

(a) Show that $\text{span}\{s_j\}_{j=1}^N = \text{span}\{u_j\}_{j=1}^K$.

(b) Let $A_i = u_i v_i^T$ be a rank-1 matrix for $1 \leq i \leq N$. Show the orthogonal decomposition $S = \sum_{i=1}^K \sigma_i A_i$ in the Frobenius inner product

$$\langle A, B\rangle_F = \sum_{i=1}^M \sum_{j=1}^N a_{ij} b_{ij},$$

i.e. show that $\langle A_i, A_j\rangle_F = \delta_{ij}$, where δ_{ij} is the Kronecker delta.

(c) For $d < K$ we may approximate S with $\sum_{i=1}^d \sigma_i A_i$; this is the POD of S. Show the error formula

$$\left\| S - \sum_{i=1}^d \sigma_i A_i \right\|_F^2 = \sum_{i=d+1}^K \sigma_i^2,$$

where $\|\cdot\|_F$ is the norm subordinate to the Frobenius inner product. This expression could be used to find the number d so that the error $\sum_{i=d+1}^K \sigma_i^2$ is within a given tolerance.

5. Consider the two-point boundary value problem

$$\mathcal{A}u := -(a(x)u')' + c(x)u = f(x) \quad \text{in } (0,1), \quad u(0) = u(1) = 0,$$

where $a(x) \geq a_0 > 0$ and $c(x) \geq 0$ are smooth functions in $[0,1]$, and $f(x) \in L^2(0,1)$.

(a) Derive the variational form for this problem.

(b) Set $a(x) = 1$ and $c(x) = 0$ and consider a partition of $[0,1]$

$$0 = x_0 < x_1 < \cdots < x_N = 1.$$

Write the system of equations produced by the finite element method with continuous piecewise linear elements $\Phi_j(x)$, $j = 1, 2, \ldots, N-1$,

$$\Phi_j(x) = \begin{cases} \frac{x - x_{j-1}}{x_j - x_{j-1}}, & x_{j-1} \leq x \leq x_j, \\ \frac{x_{j+1} - x}{x_{j+1} - x_j}, & x_j \leq x \leq x_{j+1}, \\ 0, & \text{otherwise.} \end{cases}$$

(c) Show that this system of equations is equivalent to the one produced by the finite difference method for the uniform partition of $f \equiv$ constant.

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SOLUCIÓN — PROBLEMA 5

Problema 5 — Problema de valores en la frontera: forma variacional y elementos finitos

$$\mathcal{A}u := -(a(x)u')' + c(x)u = f(x) \ \text{ en } (0,1), \qquad u(0) = u(1) = 0,$$

con $a(x) \geq a_0 > 0$, $c(x) \geq 0$ suaves, $f \in L^2(0,1)$.

(a) Forma variacional.

Sea v una función de prueba en $H_0^1(0,1) = \{v \in H^1(0,1) : v(0) = v(1) = 0\}$. Multiplicando la ecuación por v e integrando en $(0,1)$:

$$\int_0^1 \big[-(au')' + cu\big] v\,dx = \int_0^1 fv\,dx.$$

Integrando por partes el primer término:

$$\int_0^1 -(au')'v\,dx = \big[-au'v\big]_0^1 + \int_0^1 au'v'\,dx = \int_0^1 au'v'\,dx,$$

ya que el término de frontera se anula porque $v(0) = v(1) = 0$. Se obtiene la **forma variacional**: hallar $u \in H_0^1(0,1)$ tal que

$$\boxed{\mathcal{B}(u,v) := \int_0^1 a(x)u'(x)v'(x)\,dx + \int_0^1 c(x)u(x)v(x)\,dx = \int_0^1 f(x)v(x)\,dx =: F(v) \quad \forall v \in H_0^1(0,1).}$$

($\mathcal{B}$ es bilineal, simétrica, continua y coerciva en $H_0^1(0,1)$ gracias a $a \geq a_0 > 0$, $c \geq 0$ y la desigualdad de Poincaré, garantizando por Lax-Milgram existencia y unicidad de solución débil.)

(b) Sistema de elementos finitos lineales ($a \equiv 1$, $c \equiv 0$).

Con $a = 1, c = 0$: $\mathcal{B}(u,v) = \int_0^1 u'v'dx$. Sea $V_h = \operatorname{span}\{\Phi_1, \dots, \Phi_{N-1}\}$ el espacio de elementos finitos lineales continuos con Φ_j las funciones sombrero asociadas a la partición $0 = x_0 < x_1 < \cdots < x_N = 1$ (no necesariamente uniforme), $h_j := x_j - x_{j-1}$. Se busca $u_h = \sum_{j=1}^{N-1} U_j\Phi_j$ tal que $\mathcal{B}(u_h, \Phi_i) = F(\Phi_i)$ para $i = 1, \dots, N-1$ (método de Galerkin).

Como $\Phi_j'(x) = 1/h_j$ en (x_{j-1}, x_j), $\Phi_j'(x) = -1/h_{j+1}$ en (x_j, x_{j+1}), y 0 fuera de (x_{j-1}, x_{j+1}), la **matriz de rigidez** $K_{ij} = \int_0^1 \Phi_i'\Phi_j'dx$ es tridiagonal con

$$K_{jj} = \frac{1}{h_j} + \frac{1}{h_{j+1}}, \qquad K_{j,j-1} = K_{j-1,j} = -\frac{1}{h_j}, \qquad K_{ij} = 0 \text{ si } |i-j| \geq 2.$$

El sistema resultante, para $j = 1, \dots, N-1$ (con $U_0 = U_N = 0$), es

$$-\frac{1}{h_j}U_{j-1} + \left(\frac{1}{h_j} + \frac{1}{h_{j+1}}\right)U_j - \frac{1}{h_{j+1}}U_{j+1} = F_j, \qquad F_j = \int_0^1 f\Phi_j\,dx.$$

En particular, para la **partición uniforme** $h_j \equiv h = 1/N$:

$$\frac{-U_{j-1}+2U_j-U_{j+1}}{h}=F_j=\int_0^1 f\Phi_j\,dx,\qquad j=1,\dots,N-1.$$

(c) Equivalencia con diferencias finitas para $f\equiv$ constante (malla uniforme).

El esquema de diferencias finitas centradas estándar para $-u''=f$ (con f constante) en la malla uniforme es

$$\frac{-U_{j-1}+2U_j-U_{j+1}}{h^2}=f,\qquad j=1,\dots,N-1,\quad U_0=U_N=0.\tag{FD}$$

Del lado de elementos finitos, cuando f es constante:

$$F_j=\int_0^1 f\Phi_j\,dx=f\int_{x_{j-1}}^{x_{j+1}}\Phi_j\,dx=f\left(\frac{1}{2}h\cdot 1+\frac{1}{2}h\cdot 1\right)=fh,$$

pues $\int \Phi_j\ dx$ es el área bajo el "sombrero" triangular: dos triángulos de base h y altura 1, área total h. Sustituyendo en la ecuación de FEM del inciso (b):

$$\frac{-U_{j-1}+2U_j-U_{j+1}}{h}=fh\iff\frac{-U_{j-1}+2U_j-U_{j+1}}{h^2}=f,$$

que es exactamente (FD), con las mismas condiciones de frontera $U_0=U_N=0$. Como ambos son el **mismo** sistema lineal tridiagonal, tienen la misma (única) solución $\{U_j\}$: el método de elementos finitos con elementos lineales reproduce exactamente el esquema de diferencias finitas centradas de segundo orden cuando f es constante y la malla es uniforme. ■

Verificación: si además $c=0$ y se usa la fórmula de cuadratura del trapecio compuesta (exacta para f lineal/constante) para aproximar F_j en el caso general, se recupera la misma equivalencia; esta es la razón estructural por la que, para coeficientes y fuente suaves, FEM lineal y diferencias finitas centradas dan resultados muy cercanos en mallas uniformes.